Chapter 9: Hyperinflations and Unsustainable Fiscal Policy

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1 Overview

So far we have considered the determination of prices assuming that fiscal policy is sustainable, and considered local variations relative to a non-stochastic steady state. Of fundamental interest in monetary economics, however, are periods of very high inflation, i.e. hyperinflation. In this chapter we use the simple model of last chapter to consider hyperinflation.

At the heart of most accounts of hyperinflation is fiscal policy imbalance. Nevertheless, we will suggest that a useful way of thinking about historical episodes of hyperinflation is that they were the result of rapid increase in the money supply. Even if the actual source of this rapid increase is most plausibly fiscal policy imbalances, we can still model it as being consistent with sustainable fiscal policy as defined in the last section, with the additional observation that the monetary authority is reacting strongly to the budget imbalance.

After modeling hyperinflation, we consider a prominent way of modeling monetary and fiscal policy interactions via the so-called Fiscal Theory of the Price Level (FTPL). A the heart of that theory is that the fiscal policy, in the language we have defined, is unsustainable, i.e. the configuration of taxes and spending does not eliminate an explosion in debt, which in turn would imply a violation of the transversality condition of household maximization, for a given path for inflation and interest rates. In this case, it is inflation that needs to adjust to ensure a unique bounded solution. While the FTPL is useful in focusing attention on the role of fiscal policy in inflation, we will show that the same equilibrium can be characterized assuming fiscal policy is sustainable. In this alternative case, however, we assume that monetary policy directly responds to fiscal imbalances. This observational equivalence is useful to clarify what the FTPL does – and does not – say. In the language of previous chapters, it says that we can experience large inflation if monetary policy responds strongly to fiscal imbalances and if fiscal imbalances go out of control. My own view is

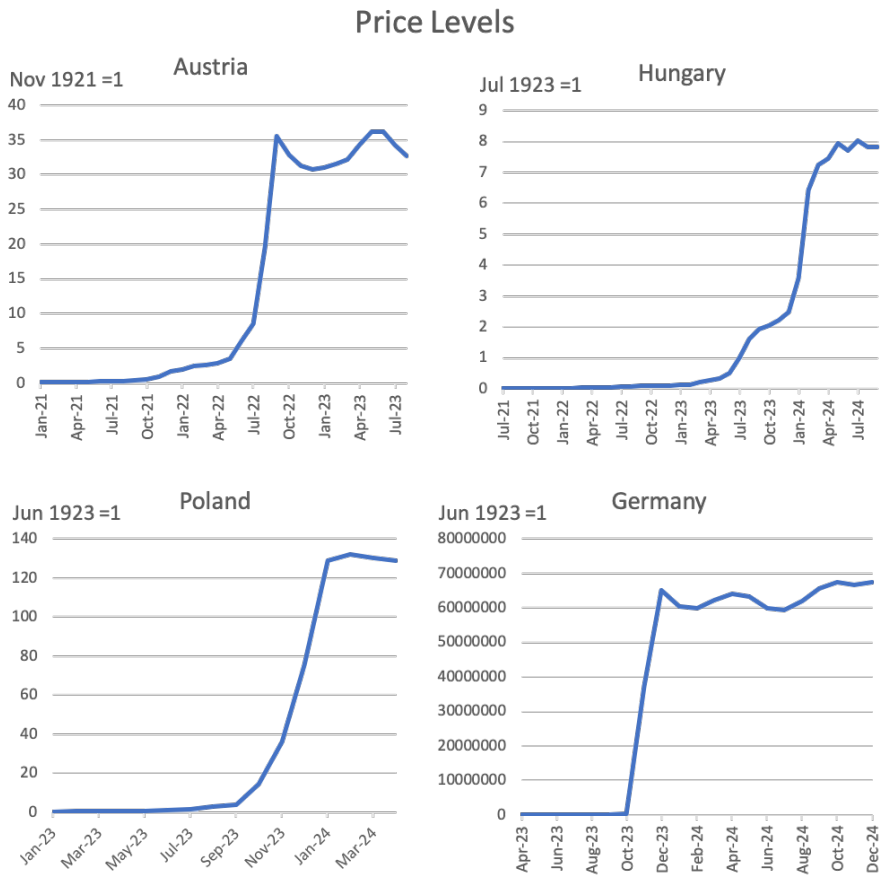


Figure 1: The price level during four hyperinflations

that assuming fiscal sustainability, but that monetary policy directly reacts to budget imbalances, is the most transparent way of modeling monetary-fiscal interactions. At the same time, I do not think there is any logical inconsistency in the FTPL way of modeling this very type of policy regime; indeed, as we will see, it produces the very same equilibrium dynamics.

2 A simple model of hyperinflation

Hyperinflation constitutes one of the most spectacular events observed in monetary economies. Here we focus on four well-known episodes of hyperinflation in Europe after World War I: the hyperinflation in Austria, Poland, Germany

and Hungary.

Figure 1 shows the price level in Austria, Poland, Germany and Hungary, normalizing it at 1 at the time the hyperinflation started. While a good that cost 1 korona in Austria in 1921 cost 35 at the end of the hyperinflation, the comparable number for the unit of account was 8 for Hungary, 130 for Poland and a staggering 60 million for Germany.

Figure 2 plots three time series for the relevant time period in the four countries: nominal money growth, inflation and real money balances. A dashed vertical line denotes the beginning and end of each hyperinflation, following Cagan's definition, which is that hyperinflation begins once inflation is above 50 percent per month and ends when inflation stays below that for a year or more. Three things immediately stand out: i) A rapid increase in the monetary base associated with ii) a rapid increase in inflation, and eventually iii) stabilization of inflation. Perhaps less obvious are the following two observations iv) inflation stabilizes *before* money growth has gone down to pre-hyperinflation levels and v) the monetary base *increases* as inflation stabilizes.

The latter two observations were, according to Sargent (1982), widely considered by contemporaries to violate the quantity theory of money. Recall that from our discussion in Chapter 4, the quantity theory of money can summarized as:

$$M_t v = Y_t P_t \tag{1}$$

where v is the velocity of money, i.e. how many transactions each unit of the currency circulates in a given period. While the quantity theory of money is merely a definition of velocity when stated as in (1), it becomes a *theory* once one makes the assumption that velocity is constant.

It is easy to see why the contemporaries cited by Sargent (1982) viewed the data from Austria as violating the quantity theory of money. The relationship in (1), for constant velocity, implies that inflation should only fall as soon as the growth rate of money stabilizes. As the figure reveals, however, the growth rate of the money supply was at around the same level when stabilization materialized as it was when hyperinflation started. Moreover, comparing high-powered money at the time when the stabilization happened in Austria, in August 1922, with high-powered money in December 1924, one finds that the money supply was 6 times higher in December 1924 than it was at the time when the stabilization occurred.

As we now will see, the model we have already sketched out is consistent with these observations because the velocity of money is endogenous. In particular, as inflation increases, the velocity of money increases. People are less willing to hold real money balances due the fact that currency is depreciating

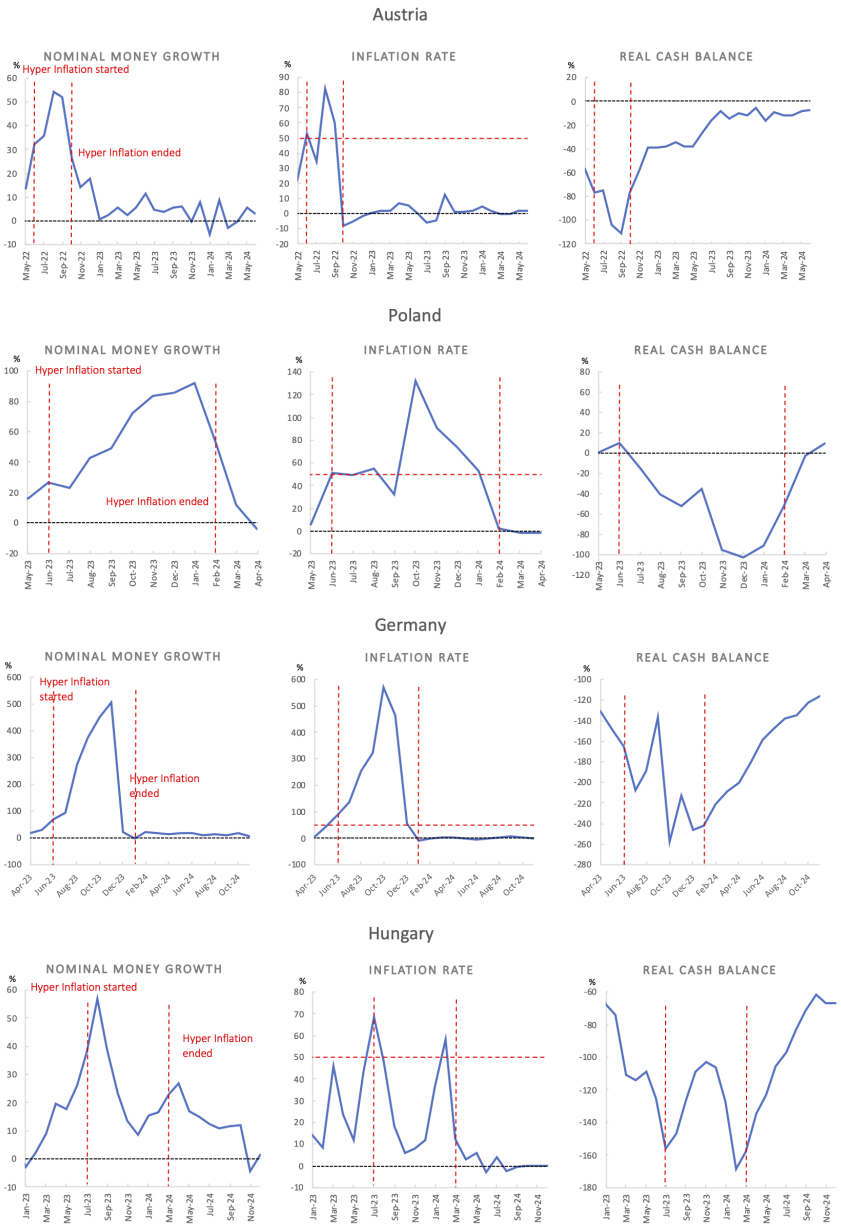


Figure 2: Four Hyperinflations

in value quickly. Moreover, it is not only the current money supply that pins down inflation, but also expectations of the future money supply. While the endogeneity of the velocity of money is important to understand the dynamics of hyperinflation, it becomes even more relevant when we think about situations in which a rapid increase in the money supply does NOT lead to inflation.

Suppose fiscal policy is sustainable in the sense we defined earlier. As we have previously shown, the basic model of price level determination can then be summarized by the following three equations:

$$\hat{m}_t = \eta_y \hat{Y}_t - \eta_i \hat{\imath}_t \tag{2}$$

$$\hat{\imath}_t = \hat{r}_t + E_t \pi_{t+1} \tag{3}$$

$$\hat{m}_t = \hat{\mu}_t + \hat{m}_{t-1} - \pi_t \tag{4}$$

which need to be augmented with a specification for monetary policy (either in terms of the interest rate or the money supply). For simplicity, suppose that that the real interest rate and output, which are exogenous in the model, are at their steady state. Hence the first and second equations can be combined to yield:

$$\hat{m}_t = -\eta_i E_t \pi_{t+1} \tag{5}$$

We assume that once hyperinflation takes place, the growth rate of the money supply takes the form:

$$\hat{\mu}_t = \beta_1 + \hat{\mu}_{t-1} \tag{6}$$

for some coefficient β_1, which is a parameter of the policy rule chosen to fit the data in the four countries.

During hyperinflation we assume that there is a probability of exiting the hyperinflationary state toward a state in which there is inflation stabilization. Once reaching this state, we assume that the money supply follows the following law of motion:

$$\hat{\mu}_t = \delta_1 \hat{\mu}_{t-1} \tag{7}$$

where $0 < \delta_1 < 1$ is a coefficient. Observe that the data from Hungary suggest that stabilization happens in two phases. After an initial stabilization, inflation breaks out again, until it is ultimately stabilized a second time. For this reason we allow for the possibility that there are two possible policy regimes that follow hyperinflation: first there is the hyperinflation regime (6), then there is a policy regime governed by (7), which we call partially credible stabilization,

where there is a probability of reverting back to the hyperinflation regime. Finally there is a "normal regime" given by (7) where there is no probability of reversal back to hyperinflation.

Conditional on being in the hyperinflation regime, the transition probability from the hyperinflation regime to the partially credible regime is α_{PC}, and to the normal regime α_N, and the probability of staying in the hyperinflation regime is then $1 - \alpha_{PC} - \alpha_N$. Conditional on being in the partially credible regime, the transition probability of going back to hyperinflation is θ_H and of going to normal θ_N, while the probability of staying in the partially credible regime is $1 - \theta_H - \theta_N$. Finally, we assume that the normal regime is an absorbing state, i.e. there is no way of moving out of it.

Under these assumptions, the model can be solved backwards. The normal regime solves (4), (5) and (7), which is solved using the method of Blanchard and Kahn.

Figure 3 shows the simulated paths for the model where $\eta_i, \beta_1, \delta_1$ and the transition probabilities have been chosen to minimize the distance between the data and the model's output.

Overall, our takeaway from these figures is that, in broad strokes, the model can account for the four observations made at the beginning of this section.

Observe that we have not explained why the money supply takes the form (6) and (7). As we already noted, this is most plausibly modeled as being due to fiscal policy. In other words the money supply follows (6) because of budget imbalances that require money printing to bridge the gap, and it transitions to (7) only once these imbalances are corrected.

3 Monetary and Fiscal Policy Interaction

This chapter is about fiscal and monetary policy interactions. On way of modeling these interactions is to assume that the fiscal policy is not sustainable, that is, that the functions $T(.)$ and $G(.)$ given V_0 for a given path of $\{Y_t, i_t, P_t, \xi_t\}$ are are not sustainable, i.e. the debt "explodes." What must happen in equilibrium is that policy becomes sustainable via inflation. Inflation helps because it both increases seigniorage revenues and reduces the real value of outstanding debt, hence requiring less taxes. This is sometimes referred as the fiscal theory of the price level (FTPL). The FTPL has been surprisingly controversial, even though at its core is the simple observation that under certain conditions fiscal policy will affect the price level. The point of this section is to suggest that the controversy around FTPL is probably overblown. In the end, there is not much difference between the FTPL and simply assuming that monetary policy



Figure 3: Four Hyperinflations Simulated

takes fiscal considerations into account (even if maintaining the technical assumption that fiscal policy is "sustainable"). Indeed, we will show that the two are observationally equivalent in an example below.

4 A Fiscal Theory of the Price Level

To illustrate the theory consider the same economy as before with a Fisher Equation:

$$\hat{\imath}_t = E_t \pi_{t+1} + \hat{r}_t$$

and a policy rule:

$$\hat{\imath}_t = \phi_\pi \pi_t$$

Consider first an equilibrium under sustainable fiscal policy in the sense we have already defined, i.e. taxes are set in such a way that the budget constraint of the government is always satisfied, so as to make sure that the transversality condition of the household is satisfied as well. As we have shown, this implies that we can study price determination independently of taxes. Government spending only matters to the extent it affects $\hat{r}_t$. For simplicity, let us suppose that $\hat{r}_t = 0$ at all times. Substitute the policy rule into the Fisher equation to obtain $\phi_\pi \pi_t = E_t \pi_{t+1}$, i.e.:

$$\pi_t = \lim_{T->\infty} (\frac{1}{\phi_\pi})^T E_t \pi_{t+T} = 0$$

If $\phi_\pi > 1$ and $\{\pi_t\}$ is a bounded process, i.e. the model has a unique bounded solution given by $\pi_t = 0$. In contrast if $\phi_\pi < 1$ the model has indeterminacy.

The fiscal theory of the price level, at least in its original form proposed by Leeper (1992), stipulates that even if $\phi_\pi < 1$ we may still be able to determine a unique bounded solution for inflation. The key condition for this to happen is that fiscal policy is not sustainable in the way we defined it before, i.e. the function $T(.)$ is no longer set so as to make sure that debt does not explode for any path of prices and interest rates. What, then, prevents debt from exploding, is that fiscal policy pins down inflation to prevent default.

Recall that the government budget constraint can be written as:

$$V_{t+1} = (1 + i_t) \left(V_t - \frac{i_t}{1 + i_t} M_t + P_t G_t - P_t T_t \right) \tag{8}$$

For this model, it is quite useful to think of the supply of money as very small so that the term $\frac{i_t}{1+i_t} M_t$ is small enough to be ignored, even if nothing changes by taking this into account. Let us define real government debt at time t as $v_t \equiv \frac{V_{t+1}}{P_t}$. We will now think about a fiscal policy in which

$$T_t = \bar{T} + \phi_\tau v_t + \bar{v} \epsilon_t^T$$

where ϵ_t^T is iid mean zero and we scale the shock by $\bar{v}$ to simplify the notation once the model is linearized. For simplicity we set $G_t = \bar{G}$. We linearize the model around a real debt $\bar{v}$. It simplifies the notation a little bit to define the fiscal variables as ratio of steady state debt $\hat{T}_t = \frac{T_t - T}{\bar{v}}$, $\hat{v}_t = \frac{v_t - \bar{v}}{\bar{v}}$, while $\hat{\imath}_t = \frac{i_t - \bar{\imath}}{1+\bar{\imath}}$ and so on, and recall that $1 + \bar{\imath} = \beta^{-1}$ in steady state. Then we can write the model as:

$$\hat{v}_t = \beta^{-1}\hat{v}_{t-1} - \beta^{-1}\pi_t + \hat{\imath}_t - \beta^{-1}\hat{T}_t$$

$$\hat{\imath}_t = E_t\pi_{t+1}$$

$$\hat{T}_t = \phi_\tau \hat{v}_t + \epsilon_t$$

$$\hat{\imath}_t = \phi_\pi \pi_t$$

Proposition 1. ***Active Monetary policy, Passive Fiscal Policy (Leeper 1992):*** *If $\phi_\pi > 1$ and $\phi_\tau > 1 - \beta$, then there is a unique bounded equilibrium in which inflation is determined only by monetary policy, and is independent of fiscal policy.*

Proposition 2. ***Active Fiscal Policy, Passive Monetary Policy (Leeper 1992):*** *If $\phi_\pi < 1$ and $\phi_\tau < 1 - \beta$, then there is a unique bounded equilibrium in which inflation is influenced by fiscal variables.*

Proposition 3. ***Passive Fiscal Policy and Passive Monetary Policy (Leeper 1992):*** *If $\phi_\pi < 1$ and $\phi_\tau > 1 - \beta$, then there is indeterminacy.*

Proposition 4. ***Active Monetary Policy and Active Fiscal Policy (Leeper 1992):*** *If $\phi_\pi > 1$ and $\phi_\tau < 1 - \beta$, then there is no equilibrium.*

To prove these propositions insert the two policy rules into the Fisher Equation and the budget constraint to yield:

$$E_t\pi_{t+1} = \phi_\pi \pi_t \tag{9}$$

$$\hat{v}_{tt} = (\beta^{-1} - \beta^{-1}\phi_\tau)\hat{v}_{t-1} + (\phi_\pi - \beta^{-1})\pi_t - \beta^{-1}\epsilon_t \tag{10}$$

$$\begin{bmatrix} E_t\pi_{t+1} \\ \hat{v}_t \end{bmatrix} = \begin{bmatrix} \phi_\pi & 0 \\ \phi_\pi - \beta^{-1} & (\beta^{-1} - \beta^{-1}\phi_\tau) \end{bmatrix} \begin{bmatrix} \pi_t \\ \hat{v}_{t-1} \end{bmatrix} + \begin{bmatrix} 0 \\ -\beta^{-1} \end{bmatrix} \epsilon_t^T$$

The eigenvalues of this matrix are given by $L(\lambda) = (\phi_\pi - \lambda)(\beta^{-1} - \beta^{-1}\phi_\tau - \lambda)$ Accordingly they are $\lambda_1 = \phi_\pi$, $\lambda_2 = \beta^{-1} - \beta^{-1}\phi_\tau$. It follows from the method of

Blanchard and Kahn that there is a unique bounded solution if one of these roots is greater than one in absolute value and one less than one, giving the conditions above. If both are less than one in absolute value, there is indeterminacy, and if both greater than one, there is no solution, as stated in the propositions above.

We have already discussed the interpretation of ϕ_π. But what about ϕ_τ? Why is there determinacy if $|\phi_\pi| < 1$, provided $\phi_\tau < 1 - \beta$? Consider the budget constraint when $\pi_t = \hat{\imath}_t = 0$ and debt is above the point around which we approximated, i.e. $\hat{v}_t > 0$. We obtain:

$$\hat{v}_t = \beta^{-1}\hat{v}_{t-1} - \beta^{-1}\hat{T}_t = \beta^{-1}(1 - \phi_\tau)\hat{v}_{t-1}$$

Hence for debt to be stable, i.e. for $\hat{v}_t$ to converge back to 0, in the absence of inflation, we need $\beta^{-1}(1 - \phi_\tau) < 1$ or $\phi_\tau > 1 - \beta$, which corresponds to taxes being higher than the interest rate cost of the debt. If on the other hand $\phi_\tau < 1 - \beta$, then debt will increase *in the absence of a change in inflation.* The reason $\phi_\tau < 1 - \beta$ leads to determinacy is that in this case, if there were no inflation, then debt would explode and there would be no unique bounded equilibrium. The way debt is stabilized is via inflation, and thus it is the budget constraint of the government that pins down the inflation rate.

$\phi_\tau < 1 - \beta$ is thus an example of **unsustainable fiscal policy**, that is, the taxing rule DOES NOT guarantee that debt does not grow without bound (and hence the transversality condition of the household is NOT satisfied for any values of $\{Y_t, i_t, P_t, \xi_t\}$, but this is how we defined sustainable policy).

We can obtain a closed form solution, for example, by using the method of undetermined coefficients (please confirm my solution below, and see if you obtained the same, Leeper does not derive closed form solutions). Provided a solution exists it will take the form, for example, for inflation and debt:

$$\pi_t = \pi^v \hat{v}_{t-1} + \pi^\epsilon \epsilon_t \tag{11}$$

$$v_t = v^v \hat{v}_{t-1} + v^\epsilon \epsilon_t \tag{12}$$

where the 4 coefficients $\pi^v, \pi^{\epsilon^T}, v^v, v^{\epsilon^T}$ are unknown. To find their values you can simply substitute this conjectured solution into (9) and (10) and match coefficients. To see how this is done, do this for equation (9). The left-hand side is given by:

$$E_t\pi_{t+1} = \pi^v \hat{v}_t + \pi^\epsilon E_t \epsilon_{t+1}^T = \pi^v v^v \hat{v}_{t-1} + \pi^v v^\epsilon \epsilon_t$$

while the right-hand side is given by:

$$\phi_\pi \pi_t = \phi_\pi \pi^v \hat{v}_{t-1} + \phi_\pi \pi^\epsilon \epsilon_t$$

so that matching coefficients between the right-hand side and the left-hand side yields:

$$\pi^v v^v = \phi_\pi \pi^v \tag{13}$$

and

$$\pi^v v^\epsilon = \phi_\pi \pi^\epsilon \tag{14}$$

Similarly for the budget constraint one obtains for the left-hand side:

$$\hat{v}_t = v^v \hat{v}_{t-1} + v^\epsilon \epsilon_t$$

and for the right-hand side:

$$(\beta^{-1} - \beta^{-1}\phi_\tau)\hat{v}_{t-1} + (\phi_\pi - \beta^{-1})\pi_t - \beta^{-1}\epsilon_t = (\beta^{-1} - \beta^{-1}\phi_\tau)\hat{v}_{t-1} + (\phi_\pi - \beta^{-1})\pi^v_{t-1}\hat{v} + (\phi_\pi - \beta^{-1})\pi^\epsilon \epsilon_t - \beta^{-1}\epsilon_t$$

so that matching coefficients between the left-hand side and right-hand side yields:

$$\beta^{-1}(1 - \phi_\tau) + (\phi_\pi - \beta^{-1})\pi^v = v^v \tag{15}$$

and

$$v^\epsilon = (\phi_\pi - \beta^{-1})\pi^\epsilon - \beta^{-1} \tag{16}$$

We can now solve for π^v, v^v, v^ϵ and π^ϵ using equations (13)-(16).

Solution for Active Monetary Policy, Passive Fiscal Policy

$\phi_\tau > 1 - \beta$ and $\phi_\pi > 1$: Observe first that one solution to equation (13) is $\pi^v = 0$ and that this solution is also consistent with the other three equations. More precisely, this solution is $\pi^v = 0$, $v^v = \beta^{-1}(1 - \phi_\tau)$, $\pi^\epsilon = 0, v^r = -\beta^{-1}$. Hence the solution is:

$$\pi_t = 0$$

$$\hat{v}_t = \beta^{-1}(1 - \phi_\tau)\hat{v}_{t-1} - \beta^{-1}\epsilon_t$$

Solution for Passive Monetary Policy, Active Fiscal Policy

$\phi_\tau < 1 - \beta$ and $\phi_\pi < 1$: Consider now the case when $\pi^v \neq 0$. Then $v^v = \phi_\pi$, $\pi^v = \frac{\phi_\pi - \beta^{-1}(1-\phi_\tau)}{\phi_\pi - \beta^{-1}}$, $\pi^\epsilon = -\frac{\phi_\pi - \beta^{-1}(1-\phi_\tau)}{(\phi_\pi - \beta^{-1})(1-\phi_\tau)}$, $v^\epsilon = -\frac{\phi_\pi}{1-\phi_\tau}$ so that the solution is:

$$\begin{aligned}
\pi_t &= \frac{\phi_\pi - \beta^{-1}(1-\phi_\tau)}{\phi_\pi - \beta^{-1}} v_{t-1} - \frac{\phi_\pi - \beta^{-1}(1-\phi_\tau)}{(\phi_\pi - \beta^{-1})(1-\phi_\tau)}\epsilon_t \\
\hat{v}_t &= \phi_\pi \hat{v}_{t-1} - \frac{\phi_\pi}{1-\phi_\tau}\epsilon_t
\end{aligned}$$

so we now see that fiscal policy directly affects inflation. It is useful to consider a special case. Suppose that $\phi_\pi = \phi_\tau = 0$. Then $\pi_t = -\epsilon_t$ and $\hat{v}_t = 0$. Hence if there is a negative tax shock the loss in tax revenues will immediately be financed via inflation. How does that work? Inflation reduces the real value of the outstanding nominal debt so that *the real value of debt* is left unchanged.

4.1 An Equivalence Result

While monetary and fiscal policy interactions are surely important, it is important to note that there is no particular reason to insist that one needs to model them in accordance to passive monetary policy and active fiscal policy just to have public debt influence inflation. Consider the following example proposed in class. Suppose that fiscal policy is sustainable in the sense we defined it earlier. In particular, assume that debt follows the following stochastic process:

$$\hat{v}_t = \rho \hat{v}_{t-1} + \epsilon_t^v \tag{17}$$

where $0 < \rho < 1$ and ϵ_t is mean zero iid. Clearly this is an example of a sustainable fiscal policy, since $\hat{v}_t$ is is a stationary process. Now assume that monetary policy follows:

$$i_t = \tilde{\phi}_\pi \pi_t - \tilde{\phi}_w \hat{v}_t \tag{18}$$

where ϕ_π, $\phi_v > 0$. This rule says that the central bank will cut the nominal interest rate in response to debt above steady state, thus taking account of fiscal condition. We can now prove the following proposition:

Proposition 5. *A policy regime with sustainable fiscal policy described by (17) and (18) is observationally equivalent to the policy regime of Passive Fiscal Policy and Active Monetary Policy for a particular value of ϕ_w. The condition for determinacy is given by $|\phi_\pi| > 1$ and is independent of fiscal variables.*

To prove this combine the Fisher equation and (18) and forward to yield:

$$\pi_t = \frac{1}{\tilde{\phi}_\pi} E_t \pi_{t+1} + \frac{\tilde{\phi}_w}{\tilde{\phi}_\pi} \hat{v}_t = (\frac{1}{\phi_\pi})^T E_t \pi_{t+T} + \frac{\tilde{\phi}_w}{\tilde{\phi}_\pi} \sum_{s=0}^{T-1} (\frac{1}{\tilde{\phi}_\pi})^s E_t \hat{v}_{t+s}$$

which yields a unique bounded solution provided that $|\phi_\pi| > 1$ given by:

$$\pi_t = \frac{\tilde{\phi}_w}{\tilde{\phi}_\pi - \rho} \hat{v}_t = \frac{\tilde{\phi}_w \rho}{\tilde{\phi}_\pi - \rho} \hat{v}_{t-1} + \frac{\tilde{\phi}_w}{\tilde{\phi}_\pi - \rho} \epsilon_t^v$$

$$\hat{v}_t = \rho \hat{v}_{t-1} + \epsilon_t^v$$

Consider a particular value for $\tilde{\phi}_\pi$, such as the traditional value $\tilde{\phi}_\pi = 1.5$ or any other value in the literature greater than 1 that generates determinacy. Then if we define $\epsilon_t^v = -\frac{\phi_\pi}{1-\phi_\tau} \epsilon_t$, $\rho = \phi_\pi$ and $\tilde{\phi}_w = (\tilde{\phi}_\pi - \rho) \frac{\phi_\pi - \beta^{-1}(1-\phi_\tau)}{\phi_\pi - \beta^{-1}}$, we get equivalent dynamics in terms of the observables $\{i_t, \pi_t, v_t\}$. This little example illustrates that there is no particular reason to move towards a model with passive monetary policy and active fiscal policy if the goal is to understand equilibrium dynamics of the model variables.

5 Do we Need a Fiscal Theory of the Price Level?